Marketing Analytics

Chapter 4 · Learning from Experiments

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Part 1 · Experiments I

Today, part B

Experiments I *(session 3)*

4.1 Back to the A/B test
4.2 The difference between two means
4.3 Testing a hypothesis
4.4 How large is the effect?
4.5 Reporting the result

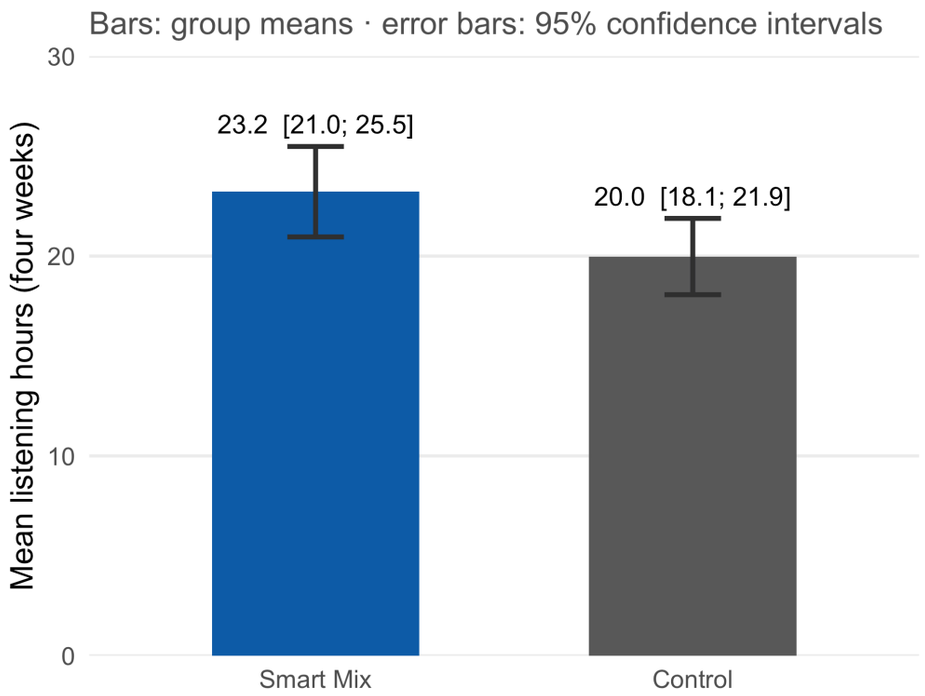
Experiments II *(session 4)*

4.6 The difference between two shares
4.7 Errors, power, and sample size
4.8 Other designs and tests
4.9 Using *p*-values responsibly

> **Driving question:** *Did it work?*

4.1 Back to the A/B test

Smart Mix: +3.2 hours




```r
t.test(hours ~ group, data = ab_test)
```

```
    Welch Two Sample t-test

data:  hours by group
t = 2.1617, df = 386.96, p-value = 0.03125
alternative hypothesis: true difference in means between group Smart Mix and group Control
is not equal to 0
95 percent confidence interval:
 0.293571 6.196096
sample estimates:
mean in group Smart Mix   mean in group Control
               23.21980                19.97497
```

Where Chapter 3 ended: two means with their **intervals** — and a test output with numbers we have not decoded yet.

Three questions for the product team

1. **Could the difference be chance?** Two random groups never have exactly the same mean — is 3.2 hours more than chance typically produces?
2. **How large is the effect — and does it matter?**
3. **Was the experiment large enough?** How large should the next one be?

> The formulas stay minimal: we compute **one** standard error by hand to see what a test does. After that, R does the arithmetic — we do the interpretation.

4.2 The difference between two means

Signal and noise

Repeat the experiment many times → the **difference** varies too. Its sampling distribution is:

1. **Centered** on the true effect
2. **Approximately normal** (central limit theorem)
3. As wide as the **standard error of the difference**:

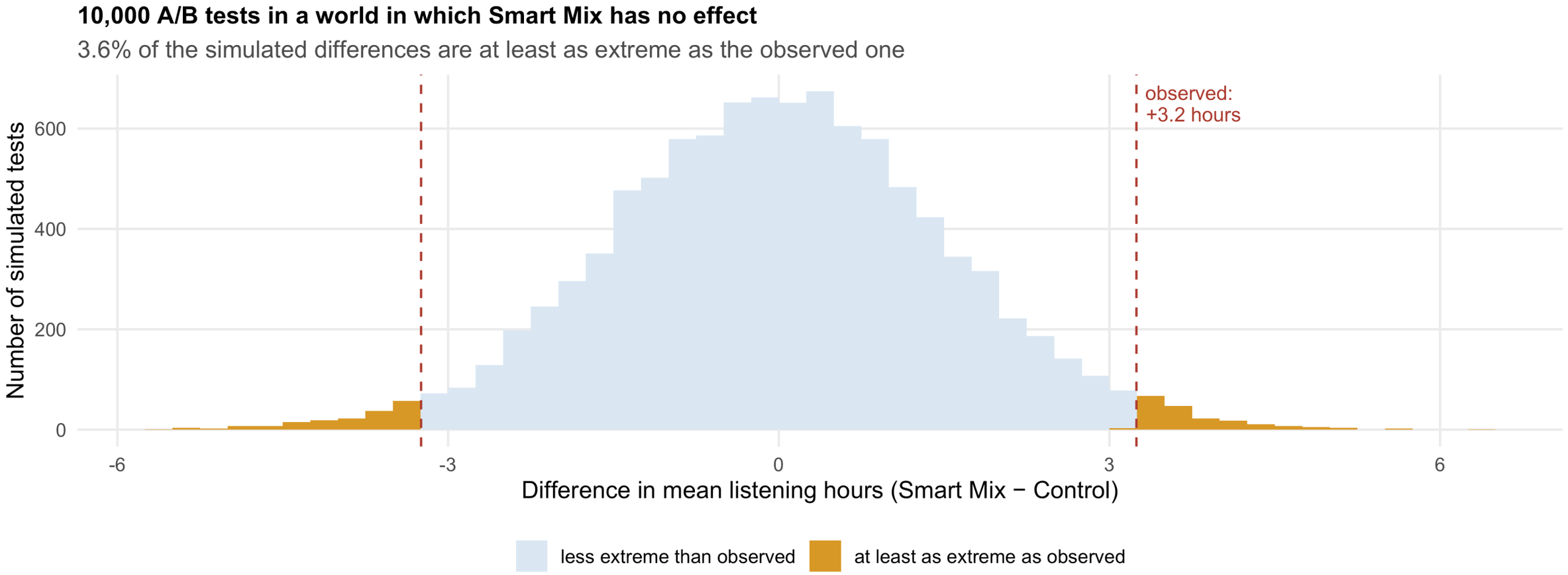
$$SE_{\bar{x}_1 - \bar{x}_2} = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

Smart Mix:

$$\sqrt{\frac{16.2^2}{200} + \frac{13.7^2}{200}} = 1.50 \text{ hours}$$

> Every test compares the **signal** (observed difference) with the **noise** (the standard error). This is the only SE we compute by hand today.

A world in which Smart Mix does nothing



If Smart Mix had **no effect**, a difference of 3.2 hours or more would occur in only **~4%** of experiments. Either Smart Mix works — or the team saw a rare accident.

Live demo: how surprising is the difference?

Course book, §4.2 — change the **users per group** and the **observed difference**.

https://wu-rds.github.io/MA2026/04-experiments.html#sec-exp-difference

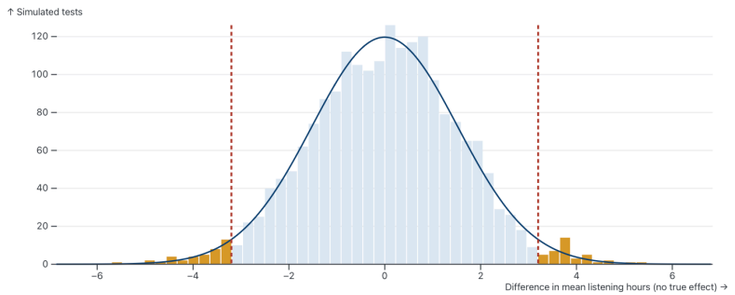
> Is a difference of 3.2 hours surprising with **50** users per group? With **800**?

Try it: how surprising is the observed difference?

The chart shows 2,000 simulated A/B tests **in a world in which Smart Mix has no effect**. The orange bars mark the tests whose difference is at least as large as the observed one. Change the **sample size per group** and the **observed difference**: when is a difference of 3.2 hours surprising — and when is it not?

Users per group 200

Observed difference (hours) 3.2



Standard error of the difference: **1.52** hours · an observed difference of 3.2 hours is **2.10** standard errors away from zero · share of no-effect tests at least as extreme: **4.0%**

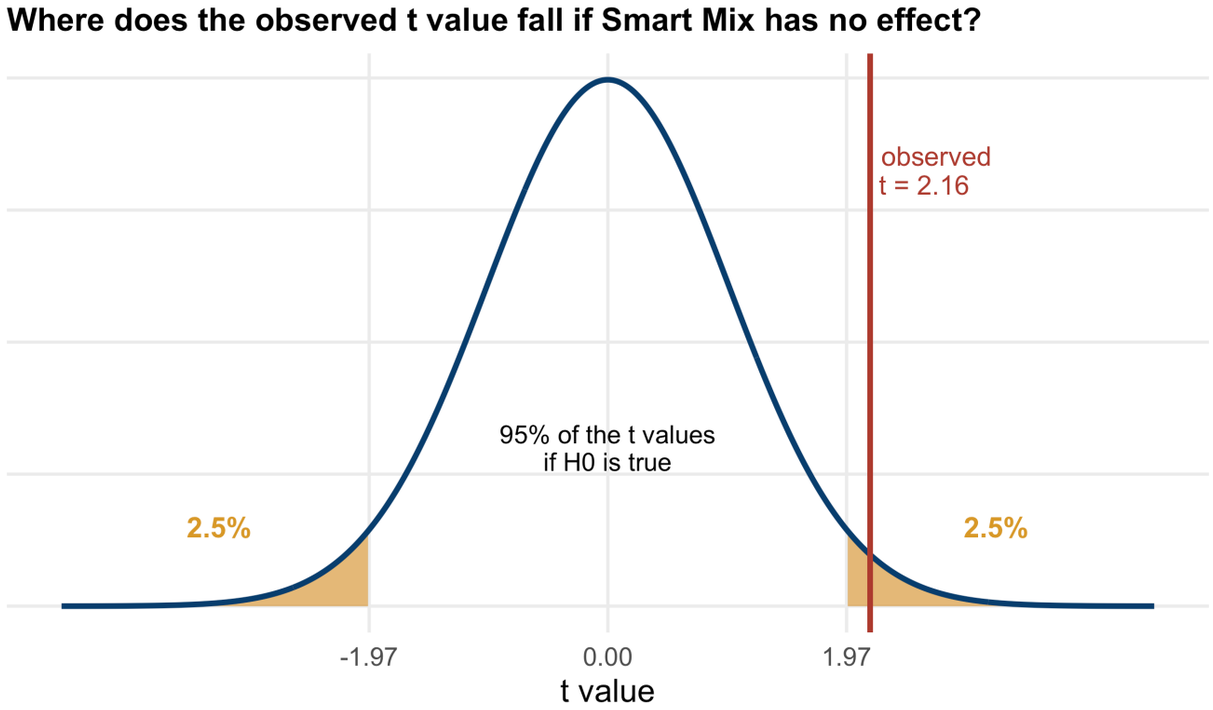
4.3 Testing a hypothesis

Null hypothesis and test statistic

- H_0: no effect, $\mu_1 = \mu_2$
- H_1: an effect, $\mu_1 \neq \mu_2$

Test statistic = signal / noise:

$$t = \frac{3.24 - 0}{1.50} = 2.16$$



If H_0 were true, 95% of all *t* values would lie between −1.97 and +1.97. Ours lies outside.

The *p*-value and the decision

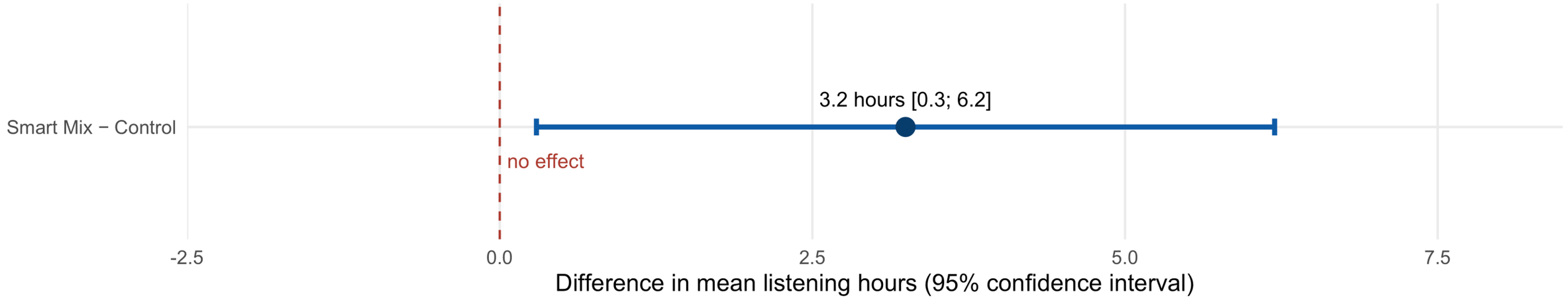
p-value: the probability of a difference at least as extreme as the observed one — **if H_0 were true**.

- Smart Mix: **p = 0.031** — in a no-effect world, ~3 in 100 experiments would show ≥ 3.2 hours
- Decision rule with significance level α = 5% (chosen **in advance**):
  - $p < \alpha \rightarrow$ **statistically significant**, reject H_0
  - $p \geq \alpha \rightarrow$ **do not reject H_0** — *not* "there is no effect"

> Our simulation said ~4%, the *t*-test says 3.1% — the test formalizes the thought experiment.

The confidence interval of the difference

$$(\bar{x}_1 - \bar{x}_2) \pm t_{0.975} \times SE_{\bar{x}_1 - \bar{x}_2} = 3.24 \pm 1.97 \times 1.50$$



CI excludes 0 ⇔ p < 0.05. Test and interval always agree.

The interval tells more: the effect could be **a quarter of an hour — or six hours**.

Reading the output

Output	Meaning
`Welch Two Sample t-test`	Two independent groups; equal variances **not** assumed (R's safe default)
`t = 2.1617`	Observed difference = 2.16 standard errors away from zero
`df = 386.96`	Degrees of freedom — shape of the *t*-distribution; irrelevant for large samples
`p-value = 0.03125`	With no effect, ≥ this difference in 3.1% of experiments
`alternative hypothesis: …`	Two-sided: a difference in **either** direction
`95 percent confidence interval: 0.29 6.20`	Effects compatible with the data
`mean in group …`	Group means; their difference = the estimated effect

Degrees of freedom — intuitively

An information budget. Each observation is one piece of information. Every quantity we **estimate** from the data uses one up. What is left — n minus the number of estimated parameters — is the information available for judging the **noise**.

Situation	Estimated	df
One sample, SD of n values	1 mean	$n - 1$
Two groups (t-test)	2 means	$n_1 + n_2 - 2$
ANOVA with 3 groups	3 means	$N - 3$
Regression with k predictors	$k + 1$ coefficients	$n - k - 1$

Why "free to vary"? Budget of 100 € for three presents, all spent: choose present 1 (30 €) and 2 (50 €) freely — present 3 is **fixed** (20 €). Three values, one constraint → **2 degrees of freedom**. Same with deviations from a mean: once the mean is known, the last deviation is determined.

> Fewer df → less information about the noise → the standard error is itself uncertain → **fatter tails**, larger critical values. Smart Mix: ~398 df → practically the normal distribution. Welch's 386.96: a weighted version of the same idea.

Your turn (5 min)

Run the test again with only the **first 50 users per group**:

```
t.test(hours ~ group,
       data = ab_test |> group_by(group) |> slice_head(n = 50))
```

How do the **difference**, the **confidence interval**, and the ***p*-value** change? Explain with the standard error.

4.4 How large is the effect?

Significant ≠ large ≠ important

- **Original unit:** +3.2 hours in four weeks
- **Relative:** +16% vs. control (20.0 hours)
- **Standardized:** Cohen's d

$$d = \frac{\bar{x}_1 - \bar{x}_2}{s_{\text{pooled}}}$$

```
psych::cohen.d(ab_test$hours, ab_test$group)$cohen.d
```

```
          lower     effect       upper
[1,] -0.4131497 -0.2167113 -0.02000216
```

(Sign = order of the groups; size ≈ 0.22)

Conventions (Cohen, 1992): 0.2 small · 0.5 medium · 0.8 large

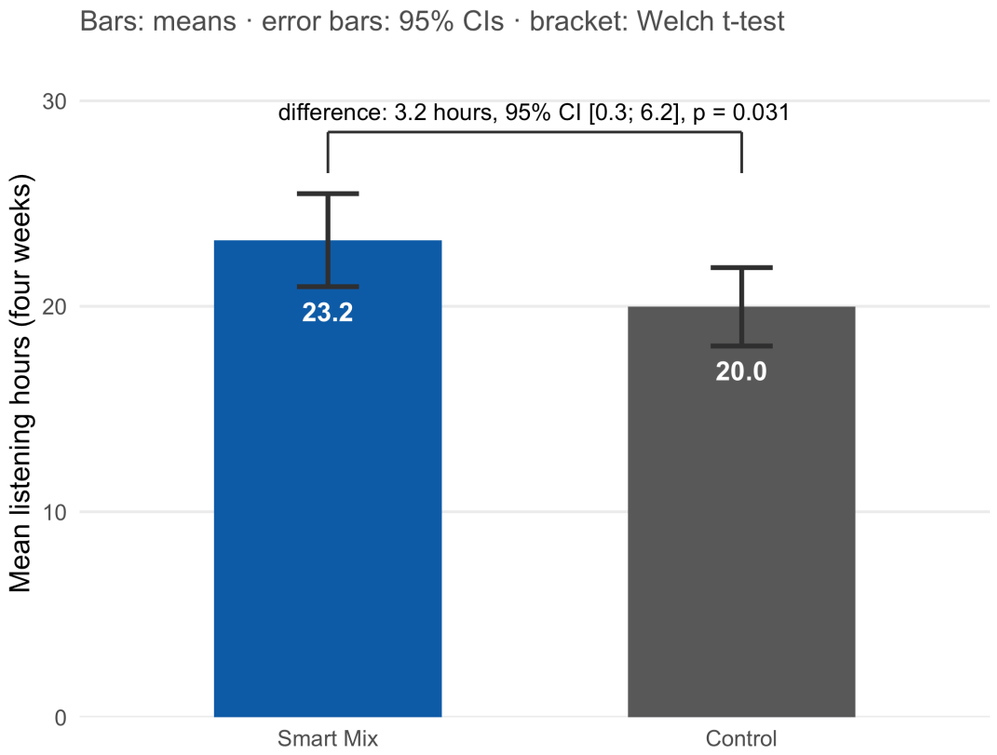
> d = 0.22: statistically **small** — but +16% listening time can be worth a lot to a service with millions of users. The conventions are a starting point, not a verdict.

4.5 Reporting the result

Effect first, uncertainty second, test last

> Users with Smart Mix listened on average 23.2 hours in four weeks, compared with 20.0 hours in the control group — an increase of 3.2 hours (+16%, 95% CI [0.3; 6.2] hours). The difference is statistically significant (Welch *t*-test: $t(387) = 2.16$, $p = .031$) but small in standardized terms ($d = 0.22$). The confidence interval is wide: the true effect could be anywhere between a few minutes and six hours.

The last sentence is the most important one: it tells the reader **how much the conclusion is worth**.



The chart that goes with the paragraph: means, their intervals, and the test result on the bracket.

Part 2 · Experiments II

Before we start

Debrief Assignment 1 *(10 min)*

- Three short paper questions about *your* results
- Closed book, no laptop

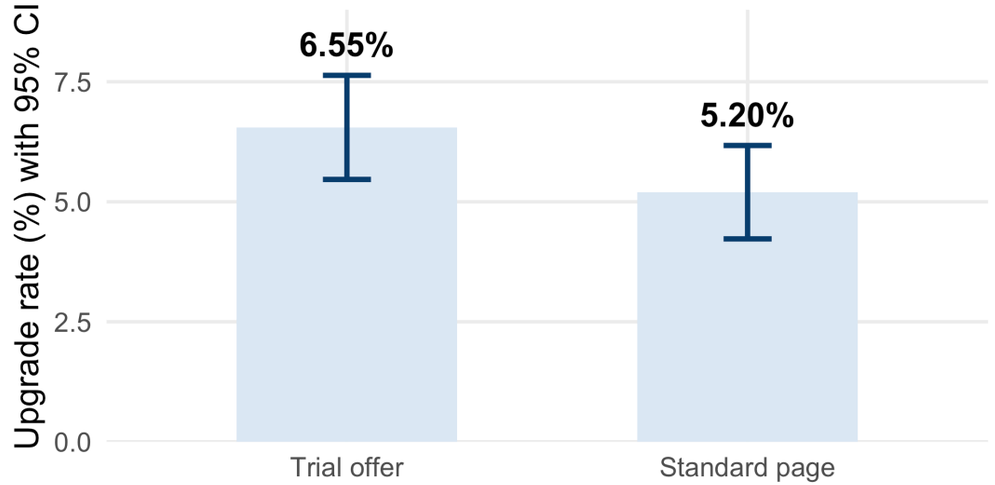
Paper quiz on Chapter 3 *(10 min)*

- Individual, closed book
- Counts toward participation

4.6 The difference between two shares

A Premium trial offer

- New Premium page with a **one-month free trial** vs. standard page
- 2,000 free users per group, randomized
- Outcome: **upgrade within two weeks** (yes/no)




```
prop.test(x = c(131, 104), n = c(2000, 2000))
```

```
    2-sample test for equality of proportions with continuity correction

data:  c(131, 104) out of c(2000, 2000)
X-squared = 3.0561, df = 1, p-value = 0.08043
alternative hypothesis: two.sided
95 percent confidence interval:
 -0.001568859  0.028568859
sample estimates:
prop 1 prop 2
0.0655 0.0520
```

Same logic as for means

$$SE_{\hat{p}_1 - \hat{p}_2} = \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}} \qquad \text{CI: } (\hat{p}_1 - \hat{p}_2) \pm 1.96 \times SE$$

- $\hat{p}(1-\hat{p})$ takes the place of s^2: it **is** the variance of a yes/no variable
- Largest at 50/50, small when (almost) everyone or no one converts

Same intuition as for means: **more variation within the groups and smaller groups → more uncertain difference.** Premium test: 1.35 ± 1.96 × 0.74 points.

“Not significant” — so it does not work?

- +1.35 percentage points (+26% relative) — but **p = 0.08**
- 95% CI of the difference: **−0.16 to +2.86** points
- Compatible with no effect — **and** with a substantial one

> The right conclusion: **“this experiment was too small to tell”** — not “the trial offer does not work”.
> (`prop.test()` = chi-square test for a 2 × 2 table; `chisq.test()` for more groups or categories.)

Your turn (5 min)

Two sales agents each talk to **300 customers**. Jon gets **200** to buy, Mark **100**.

1. Is Jon’s conversion rate significantly higher? Use `prop.test()` and report the difference with its 95% CI.
2. Does this show that Jon is the **better salesperson**?

Example after McDonnell Feit & McCullough, *Business Experiments*.

4.7 Errors, power, and sample size

Two ways to be wrong

	No effect in reality (H_0 true)	**Effect in reality (H_0 false)**
Significant	**Type I error** — false alarm (α)	✓ effect detected (**power** = $1 - \beta$)
Not significant	✓	**Type II error** — effect missed (β)

- α = 5%: one in twenty tests of a useless feature is “significant” anyway
- Power = probability of detecting a real effect; common target: **80%**
- Lower α → lower power. Fewer errors of **both** kinds → only with **larger samples**

Type I and II errors in everyday life

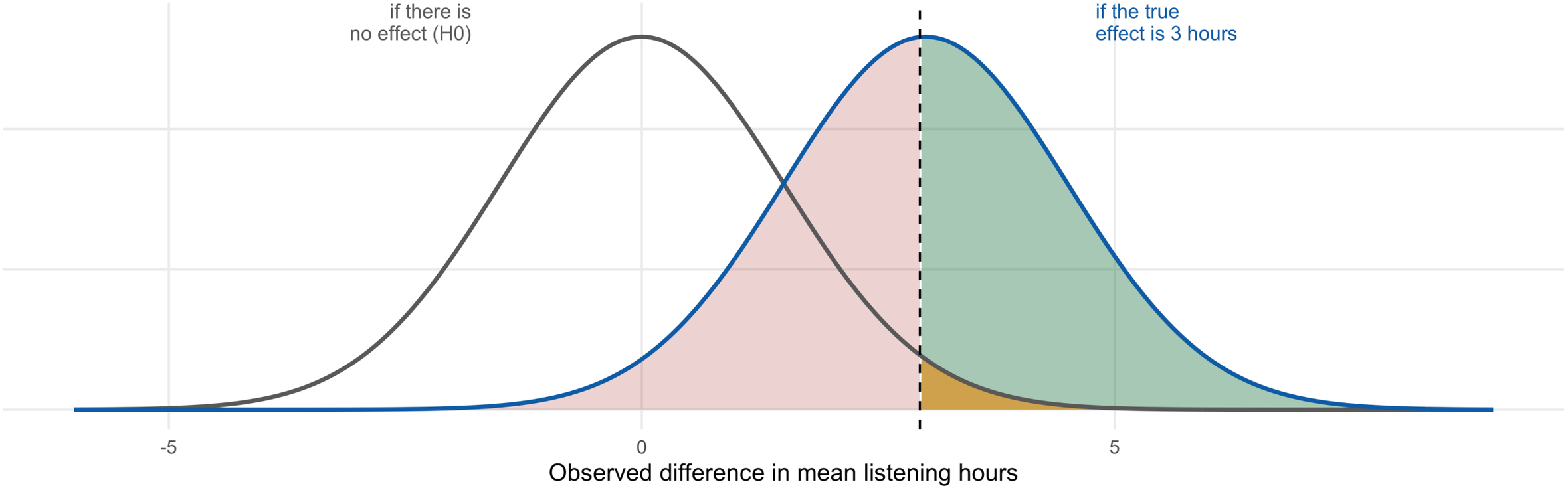
Situation	Type I error (false alarm)	Type II error (missed effect)
Pregnancy test	“Pregnant” — but she is not	“Not pregnant” — but she is
Smoke detector	Alarm because of burnt toast	Real fire, no alarm
Court trial	Innocent person convicted	Guilty person acquitted
A/B test	Useless feature rolled out	Valuable feature dropped

The court trial is a test: H_0 = “innocent” is kept unless the evidence is strong. “Not guilty” does not prove innocence — just as “not significant” does not prove “no effect”.

What determines power

Power = 52% (n = 200 per group)

Green: effect detected (power) · red: effect missed (type II error) · orange: false alarm (type I error)



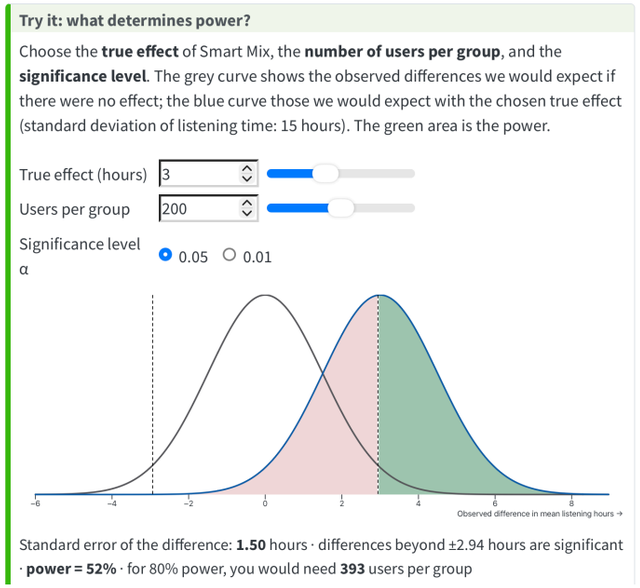
Effect size · sample size · variability. With a true effect of 3 hours, Smart Mix had only **~55% power** — the team was somewhat lucky.

Live demo: what determines power?

Course book, §4.7 — change the **true effect**, the **users per group**, and **α**.

https://wu-rds.github.io/MA2026/04-experiments.html#sec-exp-power

> Halve the true effect. How many more users do you need for the same power?



Where does the true effect come from?

In the widget, you **set** the true effect — in practice, you never know it (otherwise: no experiment needed). Power analysis uses an **assumed** effect:

- the **smallest effect that would still matter** for the decision
- evidence from **earlier experiments** (with caution)

> **No "post-hoc power":** computing power *after* the experiment with the *observed* effect adds nothing — it is just a transformation of the *p*-value. Power belongs to the **planning** stage; afterwards, look at the **confidence interval**.

Sample size: decide before you start

Means — Smart Mix, $d = 0.22$, 80% power:

```
power.t.test(delta = 0.22, sd = 1, power = 0.80)$n
```

```
[1] 325.297
```

Shares — Premium test, 5% → 6%:

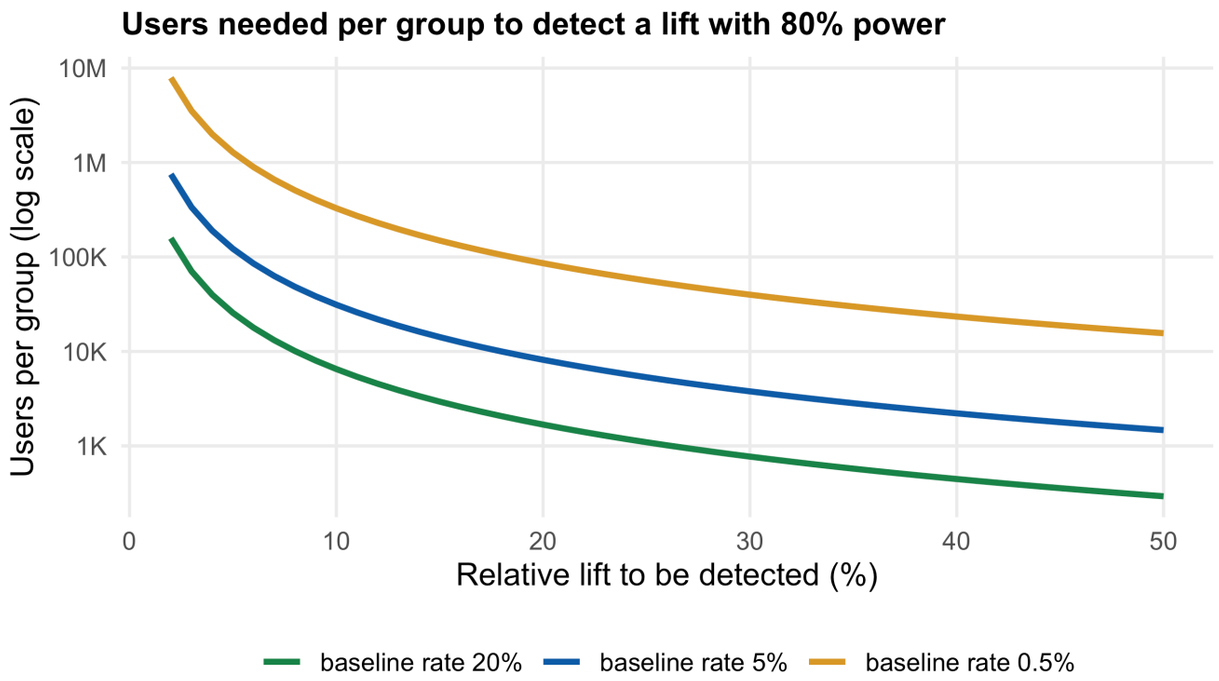
```
power.prop.test(p1 = 0.05, p2 = 0.06, power = 0.80)$n
```

```
[1] 8157.731
```

> Start from the **smallest effect that would still matter** for the decision — not from the effect you hope for.

- Smart Mix: ~330 per group (not 200)
- Premium test: **> 8,000** per group (not 2,000)
- Omit `power` and give `n` instead → the power of the test you ran

Why online advertising needs huge experiments



- CTR 0.50% → 0.55% (+10%): **> 300,000 users per group**
- Lewis & Rao (2015), *The Unfavorable Economics of Measuring the Returns to Advertising*: even experiments with **millions** of users are often too imprecise to tell whether ads pay off
- Remember eBay (§1.2): the whole US market

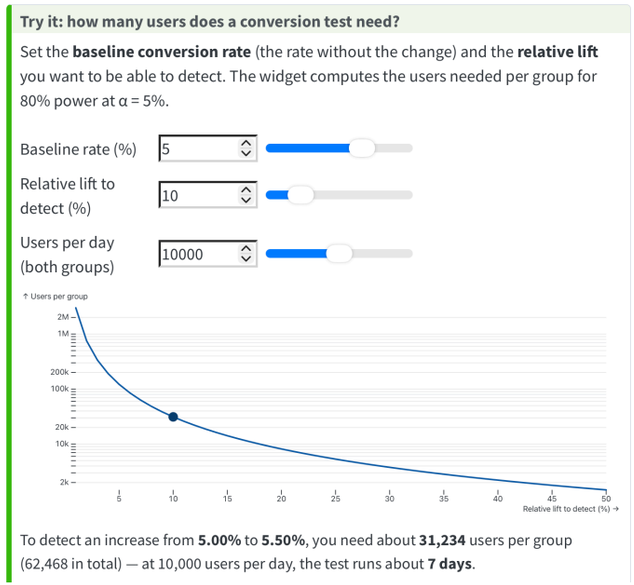
> Low baseline rates + small lifts = **enormous samples**.

Lewis & Rao (2015), "The Unfavorable Economics of Measuring the Returns to Advertising", *Quarterly Journal of Economics*, 130(4), 1941–1973.

Live demo: how many users does a conversion test need?

Course book, §4.7 — baseline rate, lift, and traffic per day.

https://wu-rds.github.io/MA2026/04-experiments.html#sec-exp-power



Your group project: ~60 per condition → only $d \gtrsim 0.5$ detectable with 80% power (small effects: < 20% power). **Make your manipulation strong.**

Your turn (5 min)

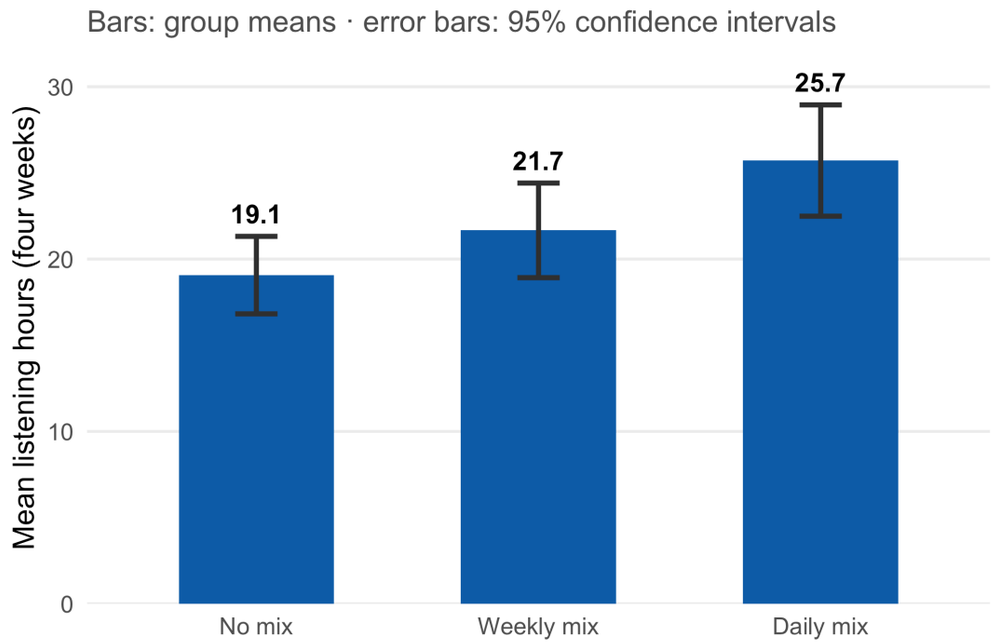
A newsletter has a click-through rate of **2%**. A new subject line should be tested.

1. How many recipients per version for detecting **+10%** (2.0% → 2.2%) with 80% power?
2. And if only **+25%** matters?

Use `power.prop.test()`.

4.8 Other designs and tests

More than two groups: why not several *t*-tests?



Three versions → three pairwise tests, each with a 5% false-alarm risk:

$$\text{family-wise error rate} = 1 - (1 - \alpha)^k$$

- 3 tests: **14%** · 10 tests (5 groups): **40%**

> **Two steps:** (1) Is there **any** difference? → ANOVA. (2) **Which** groups differ? → **corrected** pairwise comparisons.

Step 1: the ANOVA

- H_0: $\mu_1 = \mu_2 = \mu_3$ · H_1: at least one mean differs
- Signal / noise again:

$$F = \frac{\text{variation between groups (per df)}}{\text{variation within groups (per df)}}$$

- No effect → $F \approx 1$
- Effect size η^2 = share of variation explained

```r
mix_aov <- aov(hours ~ version, data = mix_test)
summary(mix_aov)
```

```
             Df Sum Sq Mean Sq F value  Pr(>F)
version       2   3367  1683.4   5.711 0.00355 **
Residuals   447 131755   294.8
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
ss <- summary(mix_aov)[[1]]$`Sum Sq`
ss[1] / sum(ss)   # eta squared
```

```
[1] 0.02491698
```

> $F(2, 447) = 5.71$, $p = .004$ — at least one version differs; but the versions explain only **2.5%** of the variation.

Step 2: corrected pairwise comparisons

```r
pairwise.t.test(mix_test$hours, mix_test$version, p.adjust.method = "bonferroni")
```

```
    Pairwise comparisons using t tests with pooled SD

data:  mix_test$hours and mix_test$version

           No mix Weekly mix
Weekly mix 0.5732 -
Daily mix  0.0026 0.1247

P value adjustment method: bonferroni
```

- **Bonferroni:** each p × number of comparisons (conservative)
- **Holm:** R's default — equally safe, more powerful
- **Tukey's HSD:** `TukeyHSD(mix_aov)` — with family-wise CIs

> Only **daily vs. no mix** differs. Unequal variances → Welch ANOVA: `oneway.test()`. The same F test appears in `lm(hours ~ version)` → Module 5.

Same people, both conditions: the design

The question: does the new search design let users find songs faster?

Within-subjects design: the **same 60 users** search for songs with the **old** design and with the **new** one, in **random order** (half old → new, half new → old). Outcome: seconds to find a song.

```
search_test <- search_data()
head(search_test, 4)
```

```
# A tibble: 4 × 3
   user old_design new_design
  <int>      <dbl>      <dbl>
1     1       38.5       39.3
2     2       38.9       30.7
3     3       43.2       50.8
4     4       51         41.8
```

- Each user appears **twice** → the two columns are **not independent**: a fast user is fast with both designs
- What varies *between* users (skill, familiarity) is huge; what we care about is the difference *within* each user
- So the test looks at **one number per user**: new − old

```
head(search_test$new_design - search_test$old_design, 8)
```

```
[1]  0.8 -8.2  7.6 -9.2  0.5  1.7 -3.7 -3.7
```

Same people, both conditions: the test

```r
t.test(search_test$new_design, search_test$old_design, paired = TRUE)
```

```
    Paired t-test

data:  search_test$new_design and search_test$old_design
t = -3.8736, df = 59, p-value = 0.0002713
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 -3.235341 -1.031326
sample estimates:
mean difference
      -2.133333
```

`paired = TRUE` tests whether the **mean of the within-user differences** is zero: the new design saves **2.1 seconds** (95% CI 1.0–3.2). Each user is **their own control** — the differences between users drop out.

```r
t.test(search_test$new_design, search_test$old_design)$p.value   #
```

```
[1] 0.1392638
```

Treating the two columns as independent groups buries the 2 seconds in the noise of how different users are: $p = 0.14$.

Non-parametric tests: ranks instead of values

- Sort all values, use their **ranks** → extremes cannot dominate
- For **ordinal** outcomes, or **strong skew / outliers in small samples**
- Ask about the **typical** observation, not the average
- Slightly less power when *t*-test assumptions hold

Design	Parametric	Non-parametric
2 groups	Welch *t*-test	Wilcoxon rank-sum
Same people	Paired *t*-test	Wilcoxon signed-rank
3+ groups	ANOVA	Kruskal–Wallis

```r
wilcox.test(hours ~ group, data = ab_test)$p.value
```

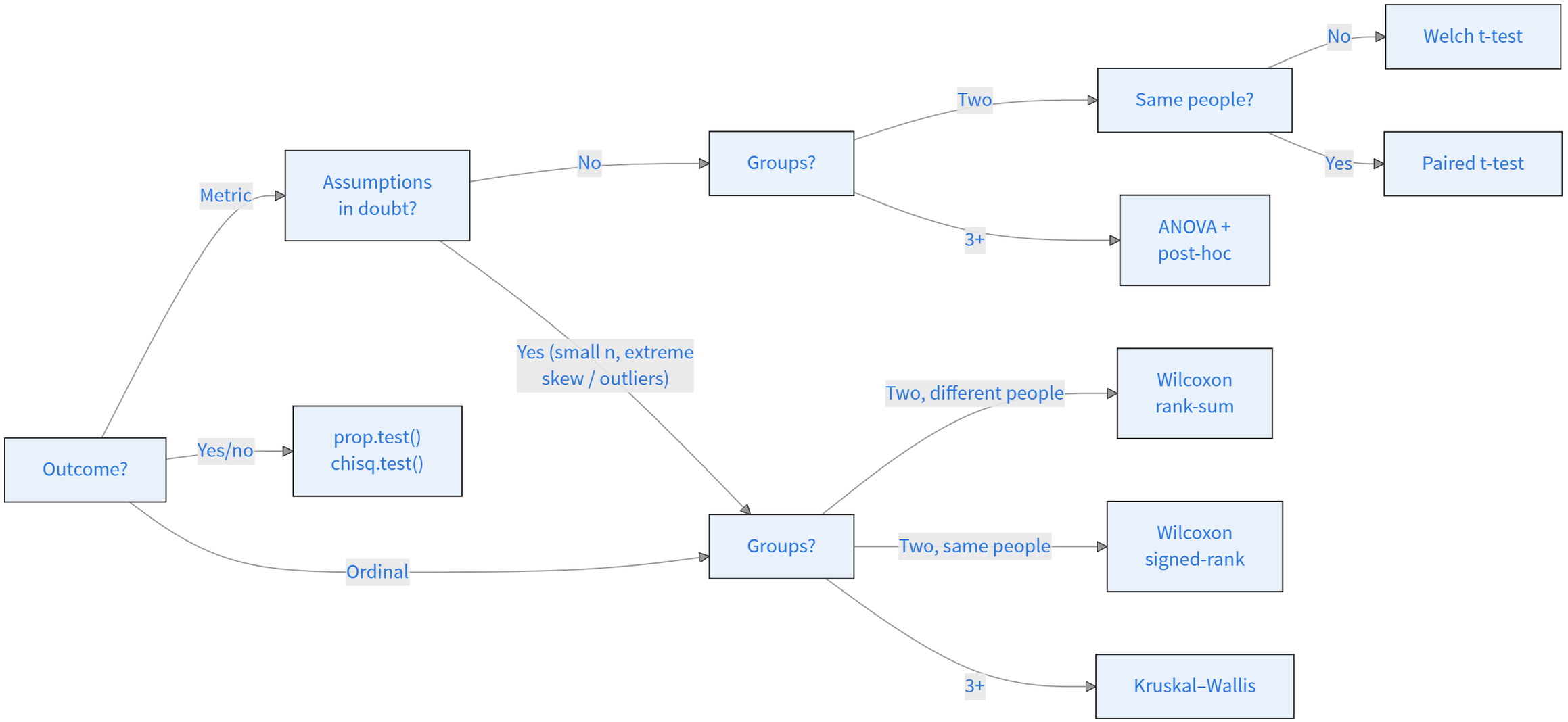
```
[1] 0.07317569
```

> Smart Mix: *t*-test $p = .031$, Wilcoxon $p = .073$ — **different questions**: average (pulled up by heavy listeners) vs. typical user. Code for all three tests: “Under the hood” boxes in the book.

Choosing a test

Outcome	Groups	Design	Test	R
Metric	2	different people	Welch *t*-test	`t.test(y ~ group)`
Yes/no	2+	different people	proportions / chi-square	`prop.test()`, `chisq.test()`
Metric	2	same people	paired *t*-test	`t.test(x1, x2, paired = TRUE)`
Metric	3+	different people	ANOVA + corrected post-hoc	`aov()`, `pairwise.t.test()`
Ordinal / extreme skew, small *n*	2	different people	Wilcoxon rank-sum	`wilcox.test(y ~ group)`
Ordinal / extreme skew, small *n*	2	same people	Wilcoxon signed-rank	`wilcox.test(x1, x2, paired = TRUE)`
Ordinal / extreme skew, small *n*	3+	different people	Kruskal–Wallis	`kruskal.test(y ~ group)`

Choosing a test: a decision tree



4.9 Using *p*-values responsibly

What a *p*-value is — and what it is not

p = P(data | H_0)

"If there were no effect, how likely would data like ours be?"

Not: P(H_0 | data)

P(runny nose | cold) ≠ P(cold | runny nose)

p = 0.031. True or false?

1. 3.1% probability that Smart Mix has no effect
2. 96.9% probability that Smart Mix works
3. The effect is important

Significant ≠ important · not significant ≠ no effect

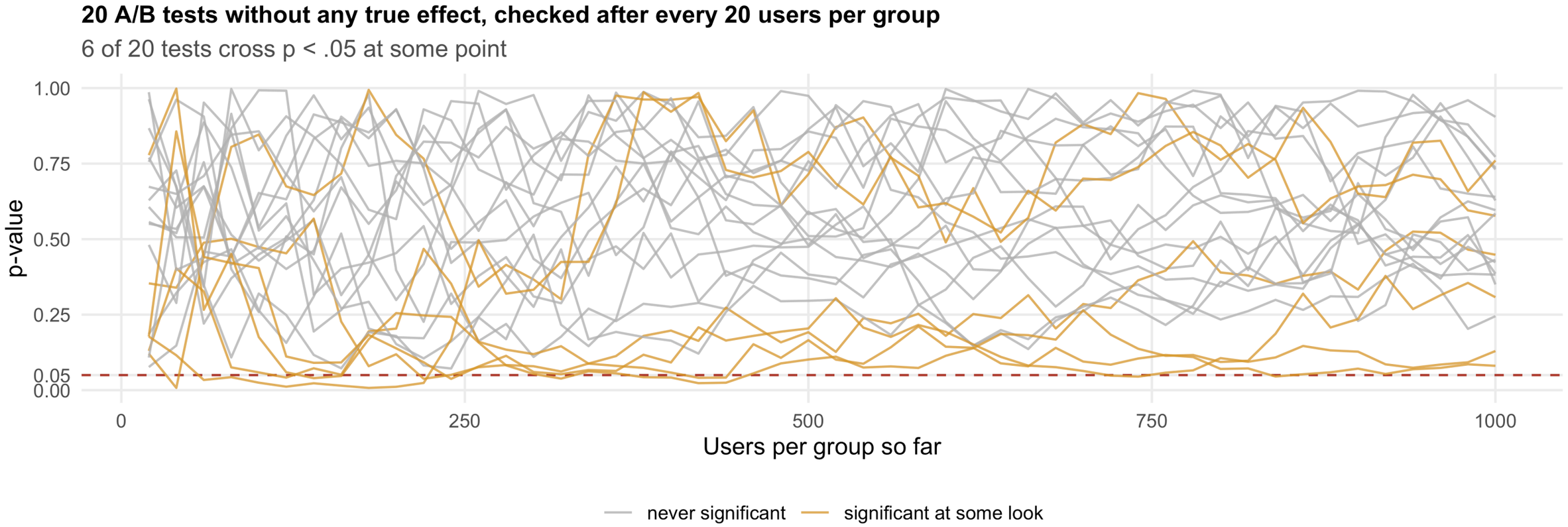
- **Significance depends on n:** with millions of users, almost everything is “significant”
- **Absence of evidence ≠ evidence of absence:** the Premium test was not significant, but compatible with +3 points

Everyday comparison	***d***	***n* per group (80%)**
Men taller than women	1.85	6
Spicy-food fans like Indian food more	0.80	26
Men weigh more than women	0.59	46
Science fans name more planets	0.07	3,669

> Nothing special happens at $p = 0.05$: 0.049 is no stronger evidence than 0.051. **Report the exact *p*, the effect size, and the CI** — as leading journals (e.g., *JM*, *JMR*) now ask authors to do.

Simmons, Nelson & Simonsohn (2013), survey of 697 US participants; Wasserstein & Lazar (2016), “The ASA Statement on p-Values”, *The American Statistician*.

Peeking destroys the *p*-value



Checking the dashboard every day and stopping at $p < 0.05$ turns a 5% false-alarm rate into **30–40%. Fix the sample size in advance; evaluate once** — or use sequential testing.

Live demo: the danger of peeking

Course book, §4.9 — 200 A/B tests **without** any effect; the analyst stops at the first $p < 0.05$.

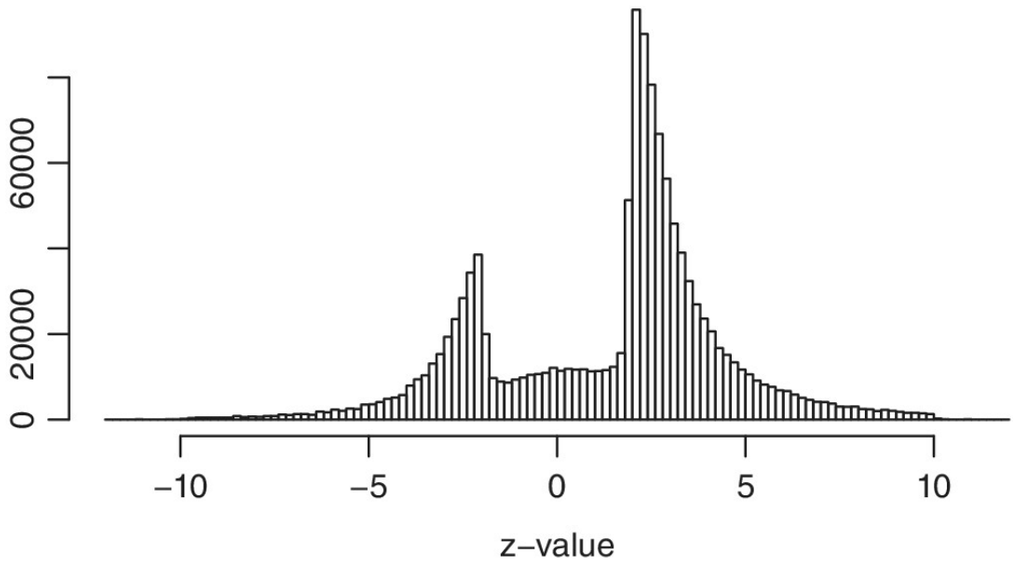
https://wu-rds.github.io/MA2026/04-experiments.html#sec-exp-nhst

> What happens to the false-alarm rate if the analyst checks **more often**, or lets the test run **longer**?



Many metrics, many segments, many tests

- 20 metrics or segments, no true effect anywhere: $1 - 0.95^{20}$ = **64%** chance of at least one “significant” result
- Reporting only the hits = **p-hacking**
- Paying attention only to “successes” = **publication bias**
- Remedy: **pre-registration** — hypotheses, outcome, *n*, and analysis recorded *before* data collection (AsPredicted, OSF); now common practice in science, incl. marketing



van Zwet & Cator (2021), *Statistica Neerlandica*, CC BY-NC-ND 4.0

Good practice for experiments

1. **Before:** outcome, hypothesis, smallest relevant effect, sample size — written down (*pre-registration*)
2. **During:** no early stopping because the result looks good (or bad)
3. **After:** effect size + confidence interval + exact *p* — planned outcome first, everything else labeled exploratory
4. **Decide** with more than the *p*-value: size, uncertainty, costs, benefits

Assignment 2: analyzing experiments

- Your **personal** experiment data (based on your student ID): an A/B test of a new onboarding flow + a three-version follow-up test
- Tasks: describe · *t*-test · effect size · shares · sample size · ANOVA
- **Deadline:** Group A Sun, Nov 8 · Group B Wed, Nov 11
- **Debrief** at the start of the following session

10 points: report 7 + debrief 3. Report effect sizes and confidence intervals — not just *p*-values. Your results may well be **non-significant** — interpret them correctly.

Learning check

1. A checkout test: +2.40 €, 95% CI [−0.30 €; 5.10 €], $p = 0.08$. Significant? One correct sentence for the head of e-commerce?
2. A new ad creative on 5 million users: CTR 0.410% → 0.415%, $p < 0.001$. Roll it out?
3. A colleague stops the test on the first day with $p < 0.05$. What is wrong?
4. Which test? (a) Conversion rates of three newsletter versions (b) Ratings (1–7) of the same 40 customers for two prototypes (c) Basket size, discount banner vs. none, 3,000 customers per group

Next session

Module 5 · Observational data I — *What if we cannot randomize?*

- Confounding, revisited — with data
- Regression as adjustment
- From correlation to (careful) causal claims